

NEMA, an American trade association, represents manufacturers of electrical equipment and medical imaging technologies. NEMA standards are particularly relevant to electrical enclosures, which define the environmental and performance requirements for enclosures used to protect electrical components from conditions like dust, water, corrosion, and physical damage. These standards help in selecting the appropriate enclosure based on the intended application and environmental conditions.

NEMA 1 defines the requirements for general-purpose electrical enclosures.

NEMA 3R enclosures are designed for outdoor use and protect against falling rain, sleet, and external ice formation.

NEMA 4 enclosures are also designed for outdoor use and provide enhanced protection against water

and dust. They offer the same level of protection as NEMA 3R enclosures but can withstand splashing water and hose-directed water.

NEMA 12 enclosures are intended for indoor use and protect against dust, dirt, and non-corrosive liquids. They are commonly used in industrial environments with excessive dust or oil presence.

Similar standards exist for various plugs and sockets. As designers, it is crucial to be aware of applicable standards and adhere to them to ensure electrical connections are compatible with other equipment.

Ergonomics

When designing products, we need to consider the abilities and variations among different people. People vary in height, and some may have difficulties operating control knobs. We must account for the force required to operate various controls.

Ergonomists design or modify work to fit the worker. An ergonomic study ensures controls are easy to operate. It takes into account the size of the human body, ensuring controls are within reach of the user without causing confusion or mistakes. It also prioritises various feedback based on their importance, ensuring proper attention is paid to the most critical items.

For example, in a car dashboard, the speedometer is given the most prominent place. Other items like fuel level, water temperature, and door conditions are given smaller spaces. This design approach ensures that while driving, the driver can quickly and unambiguously obtain information about the most critical item. Feedback and controls should be logically grouped, enabling users to quickly relate feedback to their corresponding controls. **EFY**

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